

Austin Meadows

Brooklyn, NY | austxn.meadows@gmail.com | (+1) 631 902-4991 | [personal website](#) | [GitHub](#) | [LinkedIn](#)

SUMMARY

Computational biologist and machine learning engineer finishing a **PhD (est. December 2026)** at Icahn School of Medicine at Mount Sinai and Instituto de Salud Carlos III, with a decade of wet/dry lab research across **oncology, immunology, and multi-omics genomics**. Grounded in **applied mathematics and biostatistics**: runs studies end to end, from experimental design and statistical analysis plans through quality control and publication-grade analysis of large molecular and clinical datasets. Hands-on **PyTorch and transformer engineering** with models built from scratch; builds **LLM agents** on the Anthropic API and measures them against human labels; works extensively with AI coding agents, local models, and cloud GPU infrastructure. **Evaluates frontier AI models** as an expert contributor. Co-author of papers in *Nature Communications*, *Nature Cancer*, and *Genes & Development*, including a community **benchmark of computational methods**. Focused on applying AI/ML and rigorous statistics to biomedical research and **real-world evidence**.

EXPERIENCE

Expert Contributor — Meridian AI

Jul 2026 – Present

New York, NY

- Write research-grounded reasoning challenges that expose failures in frontier AI models; scope expanded to co-developing evaluation rubrics and stumping strategies for oncology.

PhD Candidate, Biomedical Sciences & Public Health — Icahn School of Medicine at Mount Sinai

Dec 2023 – Dec 2026 (est.)

New York, NY / Madrid, Spain

- Thesis: how extracorporeal blood circulation during dialysis induces central trained immunity, in collaboration with Instituto de Salud Carlos III, using human patient samples and a mouse model profiled with RNA-seq, ATAC-seq, and CUT&Tag.
- Ran **quality control, batch correction, and sample-metadata reconciliation** for multi-omics data prepared by multiple people across separate accessioning systems.
- Authored a publication-grade statistical analysis plan; built reproducible, version-controlled Python/R pipelines covering feature prep, modeling, inferential statistics, and figures.
- Built the first ex-vivo model of dialysis-mediated trained immunity, with a diagnostic readout used within the lab.

Associate Researcher — Icahn School of Medicine at Mount Sinai (Tisch Cancer Institute)

Dec 2021 – Sep 2025

New York, NY

- Genomics and computational-biology subject-matter expert across multiple labs (Tisch, MGH, WashU), delivering analyses and tooling in a cross-institutional environment.
- Generated preliminary data underpinning successful NIAID and NCI grant applications totaling **\$4.4M**.

Sequencing Core Technician — Gihlet, Inc.

Feb 2021 – Dec 2021

New York, NY

- Built a Python interface that drives a **liquid-handling robot** to automate NGS protocols, saving tens of thousands of dollars annually.
- Processed longitudinal blood samples from a human vaccine cohort, from collection through sequencing libraries, for T-cell receptor repertoire profiling (*Mohammed et al., 2021*).

Research Technician — NYIT College of Osteopathic Medicine

Nov 2018 – Jul 2020

Long Island, NY

- Built an in-silico cell-type deconvolution method estimating tumor composition from bulk RNA-seq (*White et al., 2024*).
- Extended the CPA algorithm for single-molecule localization microscopy with a deep learning layer (CNN + autoencoder) trained on synthetic microscope data.

SKILLS

Domain Expertise: Immunology & trained immunity, cancer biology/oncology, genomics & multi-omics (RNA-seq, ATAC-seq, CUT&Tag, CUT&RUN), molecular biology, biostatistics & experimental design, survival analysis, public health

Machine Learning: PyTorch, transformers/LLMs, CNNs, GNNs, autoencoders, scikit-learn; training, fine-tuning, evaluation, hyperparameter tuning, autograd, bias/variance & validation

AI & LLM Tooling: Claude Code, OpenAI Codex, Anthropic API (agents, structured extraction); local LLMs via llama.cpp with quantization; diffusion models with LoRA adapters

Programming: Python (primary), R (tidyverse), SQL, Bash, C/C++, MATLAB

Infrastructure & MLOps: Git/GitHub, Docker, Conda, HPC (IBM LSF), GCP, AWS, cloud GPU virtual machines; CI-style reproducible pipelines

Data Engineering: Large-scale data gathering, public data mining (GEO), QC & batch correction, feature preparation, multi-omics integration, analytic dataset construction

EDUCATION

Ph.D., Biomedical Sciences & Public Health

Icahn School of Medicine at Mount Sinai / Instituto de Salud Carlos III

Dec 2026 (est.)

B.E., Biomedical Engineering • B.S., Applied Mathematics & Statistics

Stony Brook University

May 2018

SELECTED MACHINE LEARNING & AI PROJECTS *full technical write-ups at pocketmarble.github.io/projects*

- **Job-discovery agent** ([public repo](#)) — Claude, via the Anthropic API, turns a resume into a structured profile, then finds, scores, and ranks live job postings. Built an **evaluation harness against human-labeled postings** that exposed a scoring gap; fixing it halved the error.
- **iMEMORY** — Scientific advisor to an early-stage AI drug-discovery startup (Sep 2025 – May 2026) building a graph neural network over ~30,700 public human gene-expression samples to classify disease state and nominate drug targets by in-silico perturbation, with small-molecule matching and AI-proposed compounds. Consulted on platform design, biology, and target nomination; supplied PhD data; pitched to investors.
- **From-scratch builds; inspired by Andrej Karpathy's lectures:** nanoGPT (decoder-only transformer language model in PyTorch), makemore (autoregressive character-level models with training diagnostics and a hand-derived backward pass), micrograd (reverse-mode autograd engine in pure Python).

PUBLICATIONS *detailed list at pocketmarble.github.io/publications*

Mineura, K., Liu, C. R., **Meadows, A.**, et al. (2026). *Myeloid-avid mTOR-inhibiting nanobiologic attenuates allograft fibrosis after lung transplantation*. American Journal of Transplantation.

Meadows, A., Dimitrov, A., Ochando, J., et al. (2026). *Investigating the induction of trained immunity in myeloid-derived monocytes following extracorporeal circulation*. Manuscript in preparation.

White, B. S., ..., **Meadows, A.**, et al. (2024). *Community assessment of methods to deconvolve cellular composition from bulk gene expression*. Nature Communications.

Jostes, S., ..., **Meadows, A.**, et al. (2024). *H2A.Z chaperones converge on E2F target genes for melanoma cell proliferation*. Genes & Development.

Meadows, A., Sweiss, R., Kanchana, V., et al. (2024). *Active learning exercises in synaptic physiology and connectivity for the neuroscience lecture hall, laboratory course, or outreach setting*. Journal of Undergraduate Neuroscience Education.

Rialdi, A., ..., **Meadows, A.**, et al. (2023). *WNTinib is a multi-kinase inhibitor with specificity against β -catenin mutant hepatocellular carcinoma*. Nature Cancer.

Mohammed, K., **Meadows, A.**, et al. (2021). *The T cell receptor repertoire reflects the dynamics of the immune response to vaccination*. bioRxiv (preprint).

Scanga, R., Chrastecka, L., Mohammad, R., **Meadows, A.**, et al. (2018). *Click chemistry approaches to expand the repertoire of PEG-based fluorinated surfactants for droplet microfluidics*. RSC Advances.